

Jason Moon Jee

Senior Software Engineer

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SUMMARY

Innovative Software Engineer with over 12 years of specialized experience in backend development and AI technologies. Skilled in creating robust, scalable systems using Python and AWS, with a strong focus on enhancing user experiences through advanced AI models. Proven track record in deploying and managing cloud-based applications, optimizing performance, and ensuring high availability. Experienced in collaborating with diverse teams to deliver tailored solutions across multiple industries. Dedicated to continuous learning and applying the latest technologies to drive business growth and efficiency, it's me.

SKILLS

Programming Languages

Python, Java, Go/Golang, Ruby on Rails, C#/.NET, Rust, Scala, JavaScript

AI/ML

Deep Learning, Reinforcement Learning, Natural Language Processing, Computer Vision, Neural Networks, Generative AI, Prompt Engineering, Fine-tuning, Chatbot Integration, Image Synthesis

Lib & Tools

TensorFlow, PyTorch, Keras, LangChain, Haystack, Llama-index, Agentic Transformer, HuggingFace, Pandas, NumPy, OpenCV

Databases

PostgreSQL, MySQL, MongoDB, DynamoDB, Supabase, Firebase, Pinecone

MLOps

AWS, GCP, Azure, Snowflake, MLFlow, Databricks, Apache Spark, Kafka

LLM

OpenAI, Anthropic, Azure, Llama-3, SDXL, DeepSeek, Gemini, RAG, TAG, KAG

CI/CD

Git, GitLab, Jenkins, Docker, Kubernetes, CircleCI, Terraform, CDK, Pulumi

PROFESSIONAL EXPERIENCE

Zark Parking Solutions

03/2021 – Present

Senior Backend/AI Engineer

- Engineered a microservices architecture using a high-performance, lightweight **gRPC** framework with **Python/FastAPI**, and **Go(Golang)**, enabling core functionalities and **RESTful API** integration.
- Led a team to develop an ALPR system using **Python**, **OpenCV**, and **Tesseract OCR**, automating vehicle entry/exit detection with real-time license plate recognition.
- Developed a multimodal AI system combining computer vision (**OpenCV**) with **NLP-based LLMs** to process both image and text inputs enabling advanced anomaly detection, such as identifying fraudulent parking activities and preventing unauthorized vehicle access through intelligent rule-based processing.
- Enhanced accuracy with machine learning models trained on diverse datasets to improve OCR-based recognition under varying lighting and weather conditions.
- Deployed the solution on cloud infrastructure using **AWS Lambda** and **S3** for scalable, serverless storage and processing.
- Developed an intelligent chatbot using **FastAPI** and **LangChain**, integrating **OpenAI's GPT** models to provide real-time customer support, handle FAQs, and process parking-related inquiries efficiently, leveraging **Pinecone** for **Retrieval-Augmented Generation(RAG)** to enhance response accuracy and relevance.
- Implemented whisper-1 **GPT** model and **ElevenLabs API** for Voice-To-Text, Text-To-Voice functionality in the chatbot integration.
- Utilized **AWS Glue** and **Apache Spark** for large-scale data transformation, enabling real-time insights into parking trends and operational efficiency.
- Integrated **Kafka** for real-time event streaming, ensuring low-latency data ingestion and processing for ALPR-based vehicle tracking and reporting.
- Built real-time data pipeline processing 0.1M+ daily transactions with 99.99% uptime using **Databricks Delta Lake** and **Apache Spark**.
- Developed a predictive analytics model using **Python** and **Pandas** to forecast parking demand and recommend dynamic pricing strategies.
- Automated invoicing and revenue tracking by integrating **Stripe's reporting APIs**, providing real-time financial insights into parking revenue.

Blue Cross and Blue Shield of Louisiana

04/2016 – 02/2021

AI/ML Engineer

- Architected and implemented robust APIs using the **Django REST** Framework and **PostgreSQL**, which enhanced our platform's efficiency and user experience.
- Developed a dashboard to track Covid-19 spread across Louisiana, using Getis-Ord spatial analysis and Mann-Kendall trend analysis within **Azure Databricks** to pinpoint hot and cold spots at the block level.
- Built deep learning-based predictive models in **TensorFlow** and **PyTorch** to evaluate patient risk of hospitalization and emergency visits, leveraging feature extraction from claims data and ICD-10 codes.
- Applied **NLP** techniques to extract insights from the Remote Work and Covid-19 Survey for BCBSLAA.
- Emerged insights from **CRM** (Customer Relationship Management) data using Topic Modeling and Sentiment analysis.
- Utilized **Pandas** and **NumPy** for data wrangling, preprocessing, and statistical analysis, optimizing performance in ETL pipelines, time-series forecasting, and healthcare analytics.
- Implemented an AI-powered medical chatbot using **NLP (BERT, spaCy)** to assist customer support teams in handling policyholder queries, reducing response time by 50%.
- Integrated **Java** microservices with **FHIR-compliant APIs**, enabling secure data exchange between BCBSLAA and external healthcare providers.
- Deployed and managed infrastructure on **AWS**, leveraging **EC2**, **S3**, **RDS**, **Lambda**, and **API Gateway** for high-availability and cost-efficient solutions.
- Optimized claims processing workflows by leveraging reinforcement learning algorithms, improving automation efficiency and reducing manual intervention by 40%.
- Containerized applications using **Docker** and orchestrated them with **Kubernetes**, ensuring scalability, fault tolerance, and seamless **CI/CD** integration.

- Enabled streaming capabilities to facilitate real-time user interactions and developed advanced system prompts, including related image link provisioning.
- Collaborated with cross-functional teams to gather requirements, deliver tailored solutions, and conduct thorough code reviews to implement best practices in software development and deployment.

Knewton Alta

10/2012 – 03/2016

Backend Engineer

- Designed and developed scalable backend services using **Java** and **Python** to support adaptive learning platforms, improving content delivery efficiency.
- Built **RESTful** and **GraphQL APIs** to enhance data accessibility for frontend applications, integrating with **MongoDB**, **PostgreSQL**, and **Redis** to optimize query performance.
- Implemented an AI-driven recommendation engine using machine learning models (**TensorFlow**, **Scikit-learn**) to personalize learning pathways based on student performance analytics.
- Developed real-time data processing pipelines with **Apache Kafka** and **AWS Lambda**, enabling seamless event-driven learning analytics and automated content updates.
- Optimized microservices architecture for high availability and fault tolerance, leveraging **Docker**, **Kubernetes (EKS)**, and **CI/CD** pipelines for efficient deployment.
- Integrated adaptive learning algorithms that dynamically adjusted question difficulty using reinforcement learning techniques, enhancing student engagement and retention.
- Developed authentication and authorization mechanisms using **OAuth2**, **JWT**, and **SSO integrations**, ensuring secure access for students and instructors.
- Collaborated with data scientists to deploy **NLP** models that analyzed student responses, enabling AI-powered insights for educators to tailor course materials.

PROJECTS

AiPicLab.com

- Developed a service that generates high-quality children's headshots by combining facial features from father and mother's headshots using advanced generative models like **StyleGAN2**.
- Fine-tuned cutting-edge stable diffusion models using **LoRA**, **IP Adapter**, and **Textual Inversion**, significantly enhancing image output quality for end-users.
- Implemented facial feature extraction and alignment using **OpenCV** and **FaceNet**, ensuring accurate input processing for realistic image generation.
- Deployed the service on **AWS**, providing scalable infrastructure, and integrated the model into a **FastAPI** for seamless user interaction and image delivery.

Accelerometer Data for Sleep Monitoring

- **Took part in the Kaggle competition** focused on analyzing accelerometer data for sleep monitoring, aimed at detecting sleep onset and wakefulness using wrist-worn sensors.
- **Developed machine learning models** to improve large-scale sleep studies, with an emphasis on children's mental health and behavioral challenges.
- **Contributed to enhancing data-driven insights** for personalized sleep interventions and research applications.

EDUCATION

Bachelor of Science,
University of California, Berkeley

2008 – 2012 | Berkeley, US